CSE400 - CSE 400 Lecture 5 Scribe

L5_A_S2.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material for CSE 400: Fundamentals of Probability in Computing.

Use only the provided context, which consists of the lecture slides/PDF for Lecture 5 and the relevant textbook material explicitly associated with this lecture. Do not use any external knowledge and do not introduce material that does not appear in the provided context.

The lecture scribe must faithfully reconstruct what was taught in class. Include all definitions, notation, assumptions, statements of results or theorems, proofs or derivations, and worked examples exactly as they appear in the provided context. Proofs, derivations, and example solutions must be included only if they are present in the lecture or associated textbook material.

Organize the scribe clearly using section headings that follow the structure and flow of the lecture. Maintain a formal academic tone suitable for exam preparation. Ensure logical continuity so that the document can be reliably used for revision.

Do not simplify, reinterpret, or add intuition beyond what is explicitly presented in the provided context. Do not introduce new examples, explanations, commentary, or material beyond the lecture scope.

Below is a **faithful lecture scribe** reconstructed strictly from **Lecture 5 slides** and the material shown within them. No external material or added explanations have been introduced. All statements, notation, derivations, and examples follow the lecture sequence and wording as presented.

CSE 400: Fundamentals of Probability in Computing

Lecture 5 Scribe

Topic: Bayes' Theorem, Random Variables, and Probability Mass Function
Instructor: Dhaval Patel
Date: January 20, 2026

Source: Lecture slides

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1. Bayes' Theorem

1.1 Weighted Average of Conditional Probabilities

Let A and B be events. We may express

$$A = AB \cup AB^c$$

For an outcome to be in A, it must either be in both A and B or be in A but not in B.

Since AB and AB^c are mutually exclusive, by **Axiom 3**:

$$\Pr(A) = \Pr(AB) + \Pr(AB^c)$$

Using conditional probability:

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)[1 - \Pr(B)]$$

Statement:
The probability of event A is a weighted average of the conditional probabilities, with weights given by the probability of the conditioning event.

1.2 Learning by Example

Example 3.1 (Part 1)

An insurance company divides people into two classes:

- Accident prone
- Not accident prone

Statistics:

$$\Pr(\text{accident in 1 year} \mid \text{accident prone}) = 0.4$$

$$\Pr(\text{accident in 1 year} \mid \text{not accident prone}) = 0.2$$

Population assumption:

$$\Pr(\text{accident prone}) = 0.3$$

Let:

- A_1: policyholder has an accident within one year
- A: policyholder is accident prone

Then:

$$\Pr(A_1) = \Pr(A_1 \mid A)\Pr(A) + \Pr(A_1 \mid A^c)\Pr(A^c)$$

$$= (0.4)(0.3) + (0.2)(0.7) = 0.26$$

Thus:

$$\Pr(A_1) = 0.26$$

Example 3.1 (Part 2)

Given that a policyholder has an accident within a year, find:

$$\Pr(A \mid A_1)$$

Solution:

$$\Pr(A \mid A_1) = \frac{\Pr(A)\Pr(A_1 \mid A)}{\Pr(A_1)}$$

$$= \frac{(0.3)(0.4)}{0.26} = \frac{6}{13}$$

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2. Law of Total Probability and Bayes Formula

Suppose:

$$B_1, B_2, \ldots, B_n$$

are mutually exclusive events such that:

$$\bigcup_{i=1}^{n} B_i = B$$

Exactly one of the events $B_1, B_2, \ldots, B_n$ must occur.

Write:

$$A = \bigcup_{i=1}^{n} AB_i$$

Since AB_i are mutually exclusive:

$$\Pr(A) = \sum_{i=1}^{n} \Pr(AB_i)$$

Using conditional probability:

$$\Pr(A) = \sum_{i=1}^{n} \Pr(A \mid B_i) \Pr(B_i)$$

This is the **Law of Total Probability (Formula 3.4)**.

2.1 Bayes Formula

Using:

$$\Pr(AB_i) = \Pr(B_i \mid A) \Pr(A)$$

we obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{i=1}^{n} \Pr(A \mid B_i) \Pr(B_i)}$$

This is known as the **Bayes Formula (Proposition 3.1)**.

Definitions given in lecture:

- $\Pr(B_i)$: **a priori probability**
- $\Pr(B_i \mid A)$: **posteriori probability** of B_i given A

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3. Additional Example (Cards Problem)

Example 3.2

Three cards:

1. Both sides red (RR)
2. Both sides black (BB)
3. One red, one black (RB)

Procedure:

- Cards mixed
- One card randomly selected and placed down
- Observed: upper side is red

Find:

$$\Pr(\text{other side is black})$$

Let:

- RR, BB, RB denote events corresponding to card type
- R denote event that the upturned side is red

Then:

$$\Pr(RB \mid R) = \frac{\Pr(R \mid RB)\Pr(RB)}{\Pr(R \mid RR)\Pr(RR) + \Pr(R \mid RB)\Pr(RB) + \Pr(R \mid BB)\Pr(BB)}$$

Substituting values:

$$= \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)} = \frac{1}{3}$$

Hence:

$$\Pr(\text{other side black}) = \frac{1}{3}$$

The lecture further explains that assuming probability $1/2$ is incorrect because outcomes are not equally likely; considering distinct sides gives six equally likely outcomes.

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4. Random Variables

4.1 Motivation and Concept

When an experiment is performed, interest often lies in a function of the outcome rather than the outcome itself.

Examples given:

- Sum of two dice
- Number of heads in coin flips

Definition:
A random variable is a real-valued function defined on the sample space.

Formally:

$$X : \Omega \to \mathrm{R}$$

assigns to each sample point $\omega \in \Omega$ a real number $X(\omega)$.

The lecture restricts attention to **discrete random variables** (finite or countably infinite values).

4.2 Distribution of a Random Variable

Two important elements:

1. Set of values the random variable can take
2. Probabilities with which it takes those values

For any value a:

$$\{\omega \in \Omega : X(\omega) = a\}$$

is an event, and:

$$\Pr[X = a]$$

is its probability.

The collection of these probabilities is called the **distribution of the random variable**.

4.3 Visualization

Distribution may be visualized as a **bar diagram**:

- x-axis: possible values
- height: $\Pr[X = a]$

Each probability is computed from the probability of the corresponding event in the sample space.

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5. Examples of Random Variables

Example: Tossing 3 Fair Coins

Let Y = number of heads.

Possible values:

$$Y \in \{0, 1, 2, 3\}$$

Probabilities:

$$P(Y = 0) = \frac{1}{8}$$
$$P(Y = 1) = \frac{3}{8}$$
$$P(Y = 2) = \frac{3}{8}$$
$$P(Y = 3) = \frac{1}{8}$$

Since Y must take one of these values:

$$1 = \sum_{i=0}^{3} P(Y = i)$$

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6. Probability Mass Function (PMF)

6.1 Definition

A random variable that takes at most a **countable number of values** is called **discrete**.

Let X be a discrete random variable with range:

$$R_X = \{x_1, x_2, x_3, \dots\}$$

The function:

$$p(x_k) = P(X = x_k)$$

is called the **Probability Mass Function (PMF)** of X.

Property:

$$\sum_k p(x_k) = 1$$

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7. PMF Example

Given:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

Find:

- $P(X = 0)$
- $P(X > 2)$

Since:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

Thus:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Then:

$$P(X = 0) = e^{-\lambda}$$

$$P(X > 2) = 1 - P(X \leqslant 2)$$

$$= 1 - P(X = 0) - P(X = 1) - P(X = 2)$$

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8. Class Participation

Students were instructed to switch to Campuswire for participation and discussion.

If you want, I can now convert this into a **clean LaTeX-ready scribe (formatted exactly like a submission document)** so you can submit it directly.